

HYDROGENATED NBR: SIMULTANEOUS HYDROGENATION AND METATHESIS ROUTES TO LOW-MOLECULAR-WEIGHT HNBR

1. ABSTRACT

This report reviews two primary patents addressing hydrogenated nitrile rubber (HNBR) synthesis from nitrile rubber (NBR) precursors, with emphasis on low molecular weight, reduced residual unsaturation, and narrow molecular weight distribution. Both patents target the same technical problem: producing HNBR with improved processability while retaining the heat, ozone, chemical, and oil resistance associated with selective hydrogenation of acrylonitrile-butadiene rubber. The disclosed chemistry centers on a one-step, simultaneous metathesis/hydrogenation strategy rather than the more conventional two-step sequence of metathesis followed by hydrogenation. The patents describe the use of ruthenium- or osmium-based metal complexes of general formula (I) in the presence of hydrogen, optionally with a co-olefin, to convert nitrile rubber into hydrogenated nitrile rubber with lower molecular weight and lower residual double bond content. The evidence highlights that the process can deliver HNBR with Mw in the range of 20,000 to 250,000, Mooney viscosity (ML 1+4 at 100°C) in the range of 1 to 50, and polydispersity index below 3.0. Example data compare catalyst systems including Hoveyda, Arlt, and Wilkinson's catalyst, showing substantial differences in hydrogenation rate, residual double bond reduction, and final molecular weight/viscosity. The patents also reinforce the precursor composition space for NBR, including butadiene or isoprene as conjugated diene units and acrylonitrile or related unsaturated nitriles as comonomers. Overall, the landscape indicates a shift toward catalyst systems that can simultaneously cleave and saturate unsaturation in NBR, enabling controlled chain scission and hydrogenation in a single operation. This approach is technically relevant for manufacturing low-viscosity HNBR grades suitable for demanding sealing, hose, and oilfield applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber

- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a one-step process for producing hydrogenated nitrile rubber by reacting nitrile rubber in the presence of hydrogen and a ruthenium- or osmium-based compound of general formula (I), optionally with a co-olefin. The key concept is simultaneous metathesis and hydrogenation, enabling molecular weight reduction while lowering residual double bonds. - Catalyst amount: 0.041 phr — Example 1, HNBR 1 - Catalyst amount: 0.045 phr — Example 1, HNBR 2 - Catalyst amount: 0.060 phr — Example 1, HNBR 3 - Metal content: 65.6 ppm — Example 1 - Co-catalyst: NIL — Example 1, HNBR 1 - Co-catalyst: NIL — Example 1, HNBR 2 - Co-catalyst: Triphenylphosphine — Example 1, HNBR 3 - Co-catalyst amount: 1.0 phr — Example 1, HNBR 3 - Reaction time: 0 min — Example 1, Table 2 - Residual double bond (RDB): 100% — Example 1, Table 2, all HNBR grades at 0 min - Reaction time: 60 min — Example 1, Table 2 - Residual double bond (RDB): 2.5% — Example 1, Table 2, HNBR 1 at 60 min - Residual double bond (RDB): 7.0% — Example 1, Table 2, HNBR 2 at 60 min - Residual double bond (RDB): 15.4% — Example 1, Table 2, HNBR 3 at 60 min - Reaction time: 120 min — Example 1, Table 2 - Residual double bond (RDB): 0.6% — Example 1, Table 2, HNBR 1 at 120 min - Residual double bond (RDB): 4.5% — Example 1, Table 2, HNBR 2 at 120 min - Residual double bond (RDB): 3.1% — Example 1, Table 2, HNBR 3 at 120 min - Reaction time: 180 min — Example 1, Table 2 - Residual double bond (RDB): 0.5% — Example 1, Table 2, HNBR 1 at 180 min - Residual double bond (RDB): 3.9% — Example 1, Table 2, HNBR 2 at 180 min - Residual double bond (RDB): 1.3% — Example 1, Table 2, HNBR 3 at 180 min - Reaction time: 240 min — Example 1, Table 2 - Residual double bond (RDB): 0.4% — Example 1, Table 2, HNBR 1 at 240 min - Residual double bond (RDB): 3.5% — Example 1, Table 2, HNBR 2 at 240 min - Residual double bond (RDB): 0.7% — Example 1, Table 2, HNBR 3 at 240 min - Mw: 113371 g/mol — Example 1, Table 3, HNBR 1 after 240 min - Mn: 48352 g/mol — Example 1, Table 3, HNBR 1 after 240 min - PDI (Mw/Mn): 2.34 — Example 1, Table 3, HNBR 1 after 240 min - Mooney viscosity: 33.3 — Example 1, Table 3, HNBR 1 after 240 min - Mw: 133758 g/mol — Example 1, Table 3, HNBR 2 after 240 min - Mn: 54595 g/mol — Example 1, Table 3, HNBR 2 after 240 min - PDI (Mw/Mn): 2.45 — Example 1, Table 3, HNBR 2 after 240 min - Mooney viscosity: 41.3 — Example 1, Table 3, HNBR 2 after 240 min - Mw: 195304 g/mol — Example 1, Table 3, HNBR 3 after 240 min - Mn: 64130 g/mol — Example 1, Table 3, HNBR 3 after 240 min - PDI (Mw/Mn): 3.05 — Example 1, Table 3, HNBR 3 after 240 min - Mooney viscosity: 69.3 — Example 1, Table 3, HNBR 3 after 240 min - Mw range: 20,000 to 250,000 — Description - Mooney viscosity range: 1 to 50 — Description - MWD/PDI: less than 3.0 — Description

Relevant Experimental Evidence - Example 1: Compared Hoveyda, Arlt, and Wilkinson-type hydrogenation systems and showed that the inventive catalyst gave faster reduction of residual

double bonds. - Example 1: Demonstrated that the inventive catalyst systems reduced molecular weight during hydrogenation, yielding low molecular weight HNBR. - Example 1: Showed final HNBR 1 properties of Mw 113371 g/mol, PDI 2.34, and Mooney viscosity 33.3 after 240 min. - Example 1: Showed that HNBR 3, using Wilkinson's catalyst with triphenylphosphine, had higher viscosity and broader distribution than HNBR 1. - Example 1: Confirmed residual double bond reduction to 0.4% for HNBR 1 after 240 min.

Technical Relevance This patent is directly relevant because it discloses a simultaneous hydrogenation/metathesis route for converting NBR into low-molecular-weight HNBR. It provides concrete catalyst and performance data showing how catalyst choice affects residual unsaturation, molecular weight, and viscosity.

Patent 2

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses the same core process as the EP family member: reacting nitrile rubber in the presence of hydrogen and a ruthenium- or osmium-based compound of formula (I), optionally with a co-olefin, to achieve simultaneous metathesis and hydrogenation. The process is designed to produce low molecular weight HNBR with reduced residual double bonds in a single step. - Catalyst amount: 0.041 phr — Example 1, HNBR 1 - Catalyst amount: 0.045 phr — Example 1, HNBR 2 - Catalyst amount: 0.060 phr — Example 1, HNBR 3 - Metal content: 65.6 ppm — Example 1 - Co-catalyst: NIL — Example 1, HNBR 1 - Co-catalyst: NIL — Example 1, HNBR 2 - Co-catalyst: Triphenylphosphine — Example 1, HNBR 3 - Co-catalyst amount: 1.0 phr — Example 1, HNBR 3 - Reaction time: 0 min — Example 1, Table 2 - Residual double bond (RDB): 100% — Example 1, Table 2, all HNBR grades at 0 min - Reaction time: 60 min — Example 1, Table 2 - Residual double bond (RDB): 2.5% — Example 1, Table 2, HNBR 1 at 60 min - Residual double bond (RDB): 7.0% — Example 1, Table 2, HNBR 2 at 60 min - Residual double bond (RDB): 15.4% — Example 1, Table 2, HNBR 3 at 60 min - Reaction time: 120 min — Example 1, Table 2 - Residual double bond (RDB): 0.6% — Example 1, Table 2, HNBR 1 at 120 min - Residual double bond (RDB): 4.5% — Example 1, Table 2, HNBR 2 at 120 min - Residual double bond (RDB): 3.1% — Example 1, Table 2, HNBR 3 at 120 min - Reaction time: 180 min — Example 1, Table 2 - Residual double bond (RDB): 0.5% — Example 1, Table 2, HNBR 1 at 180 min - Residual double bond (RDB): 3.9% — Example 1, Table 2, HNBR 2 at 180 min - Residual double bond (RDB): 1.3% — Example 1, Table 2, HNBR 3 at 180 min - Reaction time: 240 min — Example 1, Table 2 - Residual double bond (RDB): 0.4% — Example 1, Table 2, HNBR 1 at 240 min - Residual double bond (RDB): 3.5% —

Example 1, Table 2, HNBR 2 at 240 min - Residual double bond (RDB): 0.7% — Example 1, Table 2, HNBR 3 at 240 min - Mw: 113371 g/mol — Example 1, Table 3, HNBR 1 after 240 min - Mn: 48352 g/mol — Example 1, Table 3, HNBR 1 after 240 min - PDI (Mw/Mn): 2.34 — Example 1, Table 3, HNBR 1 after 240 min - Mooney viscosity: 33.3 — Example 1, Table 3, HNBR 1 after 240 min - Mw: 133758 g/mol — Example 1, Table 3, HNBR 2 after 240 min - Mn: 54595 g/mol — Example 1, Table 3, HNBR 2 after 240 min - PDI (Mw/Mn): 2.45 — Example 1, Table 3, HNBR 2 after 240 min - Mooney viscosity: 41.3 — Example 1, Table 3, HNBR 2 after 240 min - Mw: 195304 g/mol — Example 1, Table 3, HNBR 3 after 240 min - Mn: 64130 g/mol — Example 1, Table 3, HNBR 3 after 240 min - PDI (Mw/Mn): 3.05 — Example 1, Table 3, HNBR 3 after 240 min - Mooney viscosity: 69.3 — Example 1, Table 3, HNBR 3 after 240 min - Mw range: 20,000 to 250,000 — Description - Mooney viscosity range: 1 to 50 — Description - MWD/PDI: less than 3.0 — Description

Relevant Experimental Evidence - Example 1: Compared three catalyst systems and showed the inventive catalyst reduced residual double bonds more efficiently than the industrial norm. - Example 1: Demonstrated low molecular weight HNBR formation in a single step. - Example 1: Showed HNBR 1 with Mw 113371 g/mol, PDI 2.34, and Mooney viscosity 33.3 after 240 min. - Example 1: Showed HNBR 3 with higher Mw 195304 g/mol and Mooney viscosity 69.3, indicating less effective molecular weight reduction. - Example 1: Confirmed residual double bond reduction to 0.4% for HNBR 1 after 240 min.

Technical Relevance This patent is highly relevant because it claims and exemplifies a direct route to low molecular weight HNBR from NBR using simultaneous hydrogenation and metathesis. The example data provide practical catalyst-performance comparisons that are useful for selecting conditions to control unsaturation and viscosity.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Both patents disclose the same core synthesis concept: simultaneous metathesis and hydrogenation of nitrile rubber in the presence of hydrogen using a ruthenium/osmium-based catalyst system.
- Both examples compare Hoveyda, Arlt, and Wilkinson-type catalyst systems; the inventive catalyst gives the fastest residual double bond reduction and the lowest final RDB values.
- The disclosed performance window is consistent across both patents: low residual double bonds, Mw in the low-to-mid 10^5 g/mol range in Example 1, and Mooney viscosity reduction relative to the industrial norm.
- Example 1 in both patents reports catalyst loadings of 0.041, 0.045, and 0.060 phr, with metal content fixed at 65.6 ppm.
- The industrial-norm system includes Wilkinson's catalyst with 1.0 phr triphenylphosphine co-catalyst, and it gives higher viscosity and broader PDI than the inventive catalyst systems.

- Both patents explicitly frame the process as a single-step alternative to prior art two-step metathesis-then-hydrogenation routes.

4. CONCLUSION

The evidence supports a single-step simultaneous metathesis/hydrogenation strategy for converting NBR into low-molecular-weight HNBR with low residual unsaturation. The most favorable disclosed conditions in the examples use a ruthenium-based catalyst system at 0.041 phr and 65.6 ppm metal content, with no co-catalyst, achieving RDB reduction to 0.4% after 240 min and a final Mw of 113371 g/mol with Mooney viscosity 33.3. Across both patents, the key technical implication is that catalyst selection strongly controls hydrogenation rate, chain scission, and final rheology. Based on the evidence, the recommended synthesis direction is a one-pot hydrogenation/metathesis process using a Ru- or Os-based catalyst of formula (I), with careful tuning of catalyst loading and co-catalyst use to balance low residual double bonds against excessive molecular weight growth or viscosity.

5. REFERENCES

- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>